15. STRUCTURING OF THE VENOUS SYSTEM 1 : THE CARDINAL VEINS

DURATION : 12:46 STUDY SHEET

COURSE SUMMARY

This video offers a detailed teaching on the structuring of the venous system, with a particular emphasis on the cardinal veins, which play a fundamental role in embryology. The main venous systems, their embryological origins, and the transformation of venous structures, such as the umbilical and vitelline veins, are discussed. The exploration also includes the genesis of the heart and great vessels, as well as the importance of the subcardinal and supracardinal veins in the development of the venous system. This knowledge is crucial for practitioners and students in embryology and osteopathy, enriching their understanding of the interrelationships between embryological development and vascular anatomy.

STRUCTURING OF THE VENOUS SYSTEM: THE CARDINAL VEINS

THE THREE GENERAL VENOUS SYSTEMS

Three main **venous systems** are identified within the organism:

- **The neurocranial system:** Corresponds to the vertebral plexus.
- **The viscerocranial system:** Linked to the portal system.
- **The internal system (internal man):** Represented by the inferior and superior vena cava system.

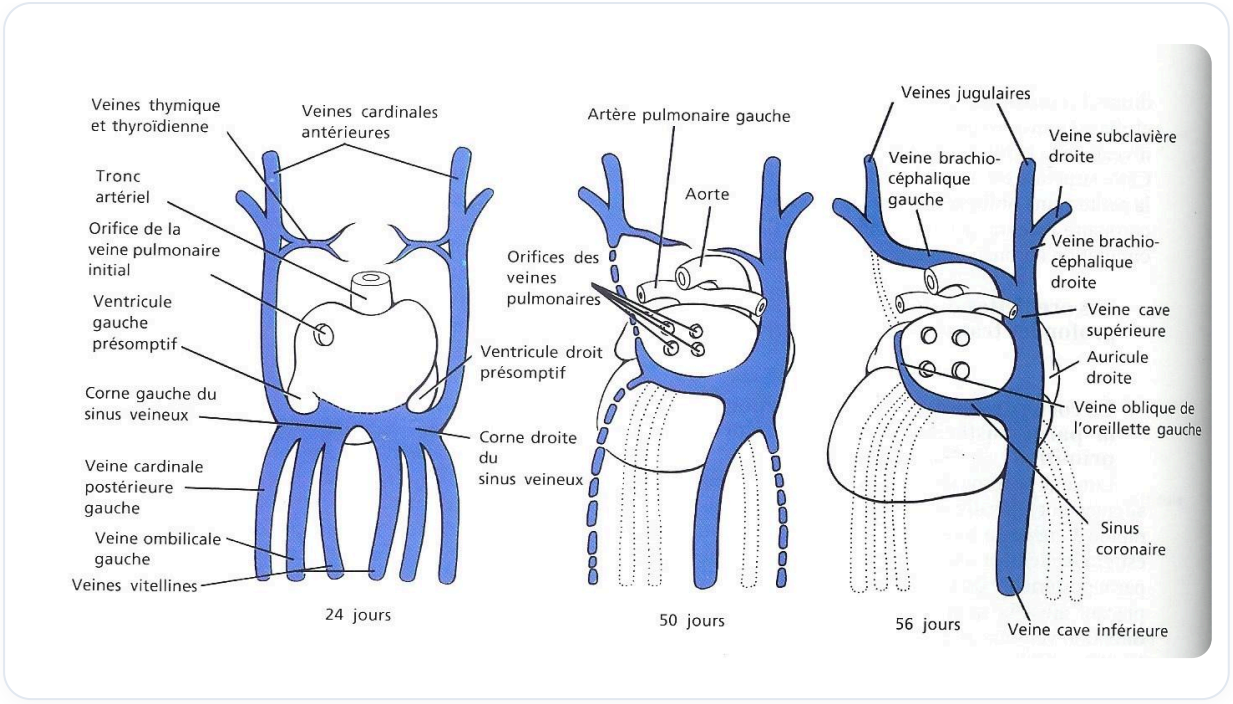


FIG. — Heart venous system development

EMBRYOLOGICAL ORIGINS OF THE VENOUS SYSTEMS

From an **embryological** point of view, the formation of these systems is linked to three precursors:

- **The common cardinal:** Gives rise to the neurocranium, the vertebral plexus, and the cava system (inferior and superior).
- **The vitelline:** Essentially linked to the viscerocranium.
- **The umbilical:** Largely disappears, leaving a terminal fibrous system. A persistent terminal vessel can form **Meckel's diverticulum**, a small pouch located just before the ileocecal valve, which can be the cause of diverticulitis.

THE CARDINAL SYSTEM

The cardinal system is divided into:

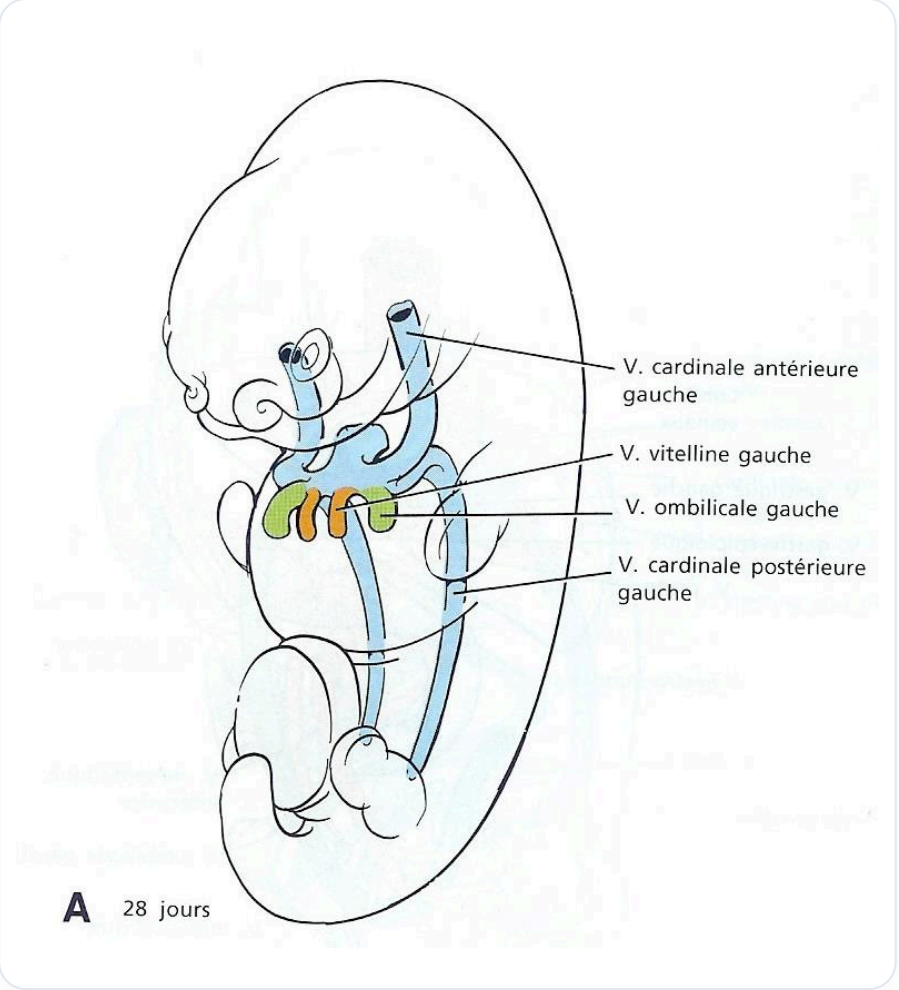


FIG. — Embryonic venous system

- **Superior or anterior cardinal vein:** Located laterally superiorly.
- **Posterior or inferior cardinal vein (or mesonephrotic):** Located laterally inferiorly.

These cardinal veins are also called **mesometric veins**. The whole corresponds to what is called the **venous sinus**. The umbilical vein and the vitelline vein are located in the middle.

FATE OF THE CARDINAL, UMBILICAL, AND VITELLINE VEINS

- **Umbilical veins:** Disappear, leaving only vestiges.
- **Right cardinal vein:** The part between the sinus and the thymic vein disappears. However, the right thymic vein anastomoses with the left thymic vein to form the **brachiocephalic vein**.
- **Vitelline vein:** The left part disappears. The right part remains and anastomoses with the right anterior cardinal vein to form the **inferior vena cava**.

THE COMPONENTS OF THE VESSEL AND CHANGES IN TRAJECTORY

For a vessel to exist, four components are necessary:

1. The **trajectory**.

2. The **metabolic concentration**.

3. The **vacuolisation**.

4. The **interstitial fluid**.

The disappearance of part of a vein indicates the loss of one of these components. A new trajectory is then drawn. Vascular development occurs laterally towards the centre and segmentally.

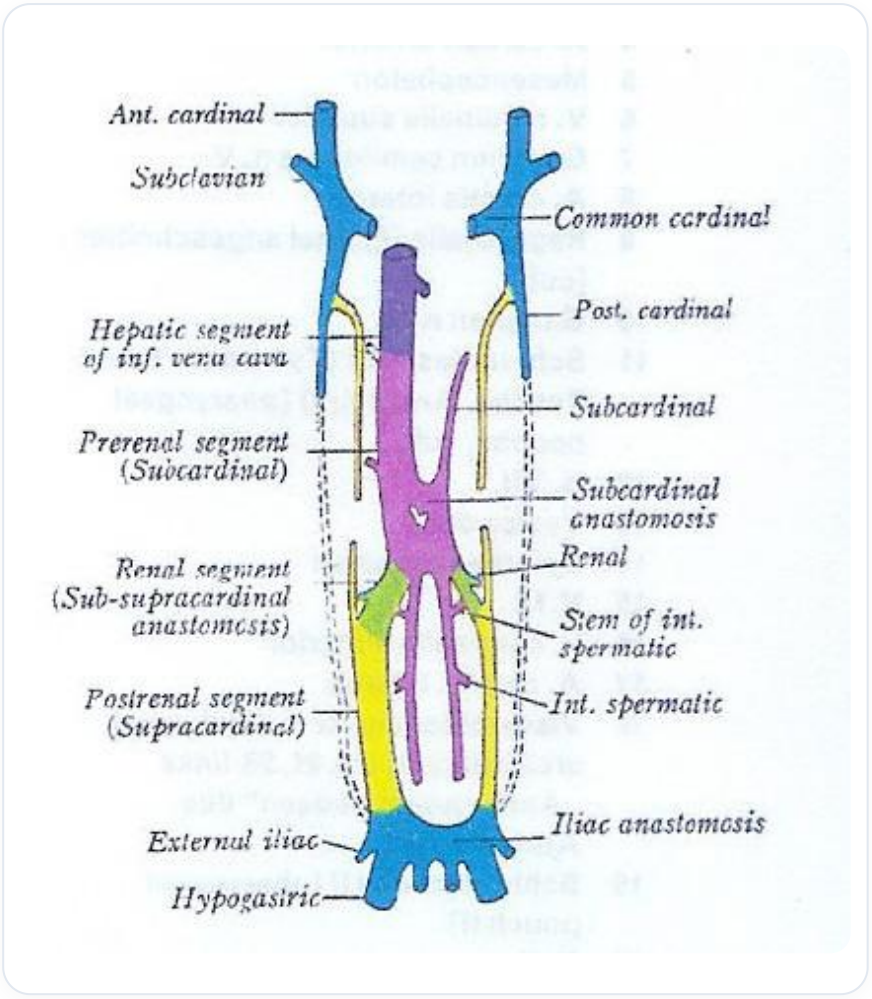


FIG. — Embryonic SVC and IVC

The vascular system of the internal man is characterised by the common cardinal vein and the posterior cardinal vein, resulting from phenomena of assimilation and disassimilation.

DEVELOPMENT OF THE HEART AND GREAT VESSELS

The heart develops from the **primitive streak**, relying on a movement of concentration towards the midline. This process includes the **corrosion field**, the **heart looping**, and its positioning towards the left. A longitudinal concentration axis is followed by the looping, creating a perpetual movement of the heart. Then follows the more horizontal positioning of the great vessels, particularly pulmonary and aortic.

The cardinal system develops more posteriorly, while the umbilical and vitelline systems are more anterior.

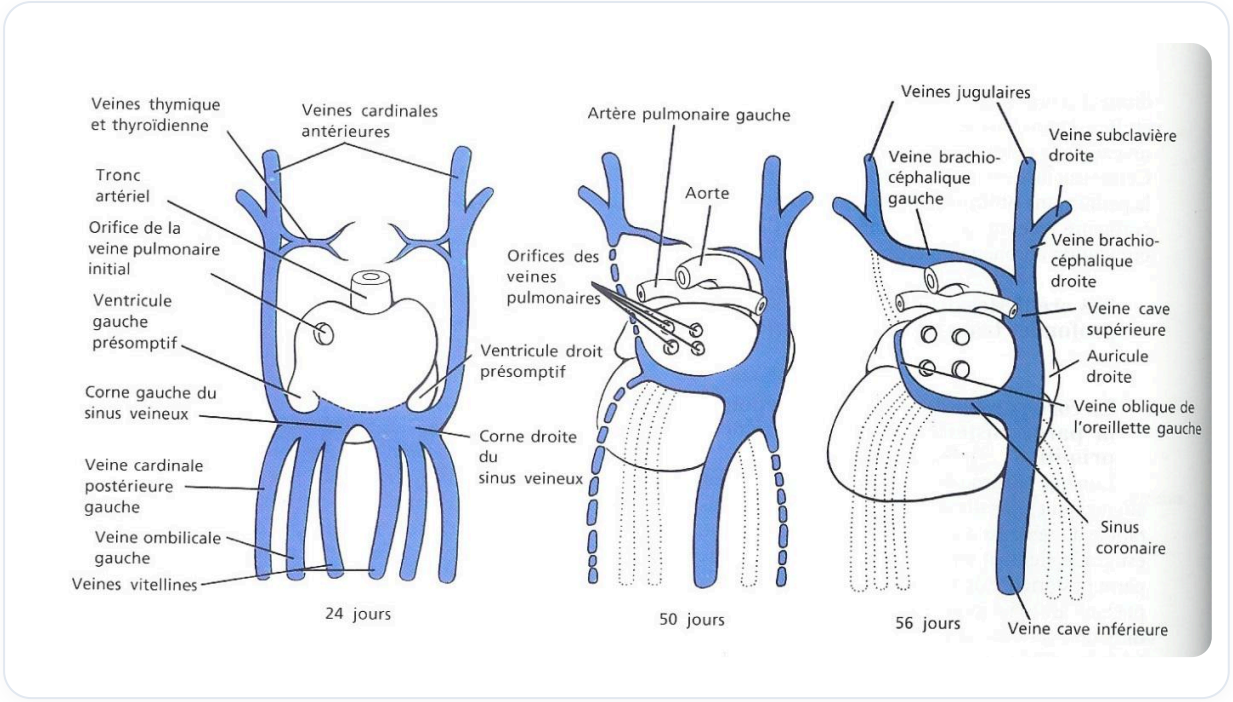


FIG. — Cardiac embryonic development

THE SUBCARDINAL AND SUPRACARDINAL VEINS

From the cardinal network, the **subcardinal veins** emerge at the base of the posterior cardinal veins.

- The major part of the left subcardinal vein regresses.
- The right subcardinal vein loses its connection with the posterior cardinal vein and develops a new anastomosis with the vitelline vein, participating in the formation of the cava system. This part will form the **inferior vena cava**, located between the liver and the kidney.

Derivatives of these subcardinals include the **suprarenal** vein, the **testicular** veins, the left renal part of the inferior vena cava, and the supracardinal.

This process is a "budding" by an aspiration field, creating new segmental trajectories while others disappear. The subcardinals re-bud to form a second network, and the process repeats with the supracardinals.

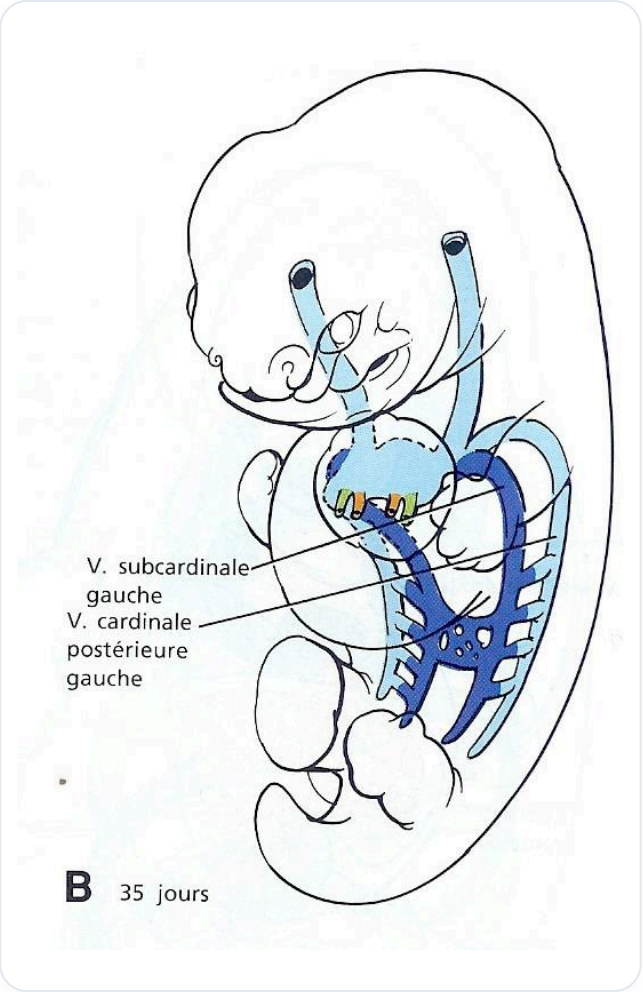


FIG. — Embryo 35 days vascularization

THE FURTHER FATE OF THE VENOUS NETWORKS

The only remaining vestige of the cardinal system will eventually be the **azygos system**. It will be replaced by the subcardinal, then supracardinal systems.

- Inferiorly, the iliac system is initially cardinal, but this system disappears.
- The subcardinal, partly disappeared, contributes to the purple segment that forms the inferior vena cava. This vein is a combination of part of the subcardinal and part of the supracardinal.

This development occurs according to a transverse movement. As a result, some segments of the inferior vena cava come from the supracardinal, with some vestiges of the subcardinal.

The developmental movement of the subcardinal is characterised by "spotting", a segmental budding that contains the entire network of the system.

DRAINAGE OF DIFFERENT AREAS

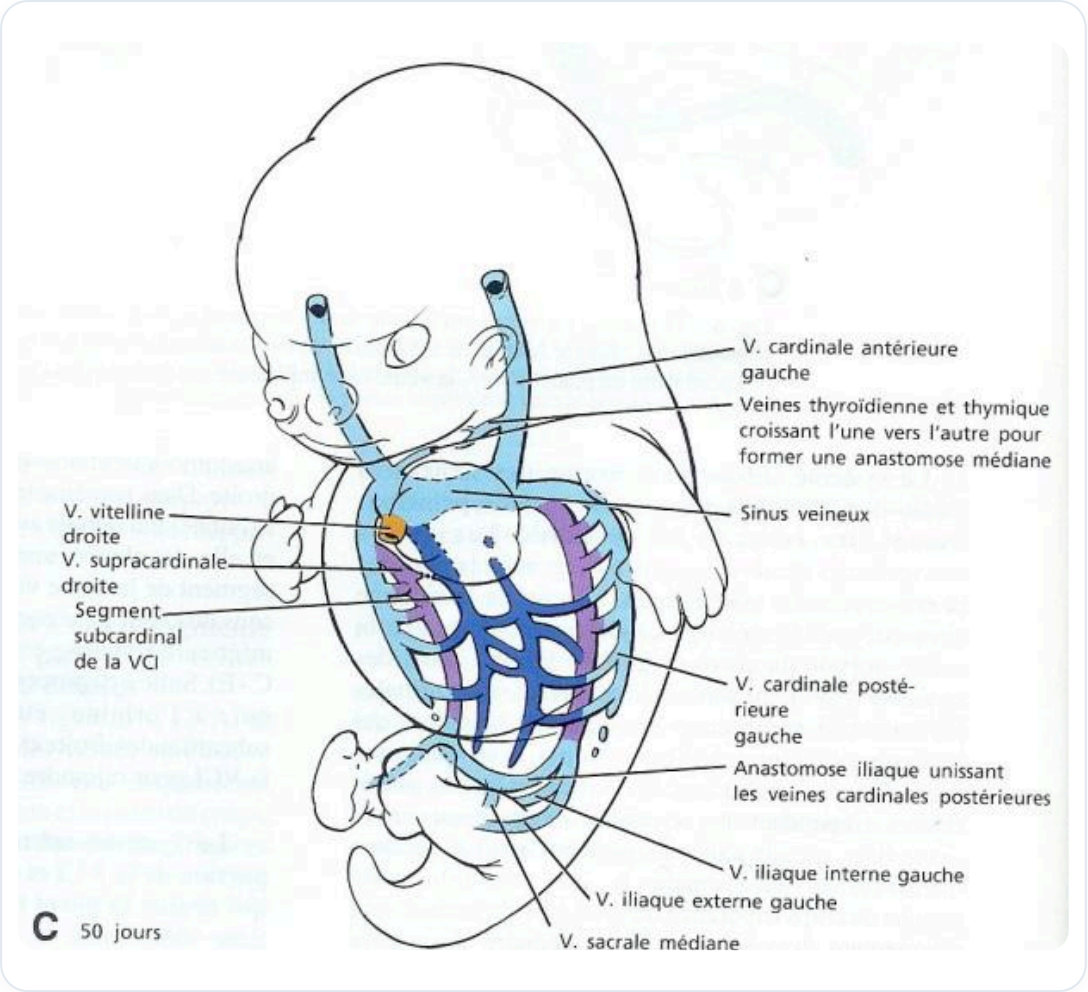


FIG. — Embryonic venous system

- **Azygos-hemiazygos system:** Ensures all thoracic drainage.
- **Mesodermal system (internal man):** Includes renal, spermatic, and suprarenal drainage.

The movement from the periphery to the centre is clearly visible. The supracardinal gives the entire thoracic and abdominal part, with a segmental difference for the left side (hemiazygos) and the right side (azygos).

The fusion of these networks in the abdominal region forms the **inferior vena cava**, which drains the entire system of the mesodermal man (support tissue).

KEY RESIDUAL VENOUS STRUCTURES

The only final vestiges are:

- **The brachiocephalic trunk.**
- **The superior vena cava.**
- **The iliac veins** below.
- **In the centre:** The meeting on the cava plane with renal, suprarenal (mesodermal type) and azygos structures.

- **To the left and right:** Hemiazygos and azygos, ensuring thoraco-abdominal drainage on the mesodermal plane.

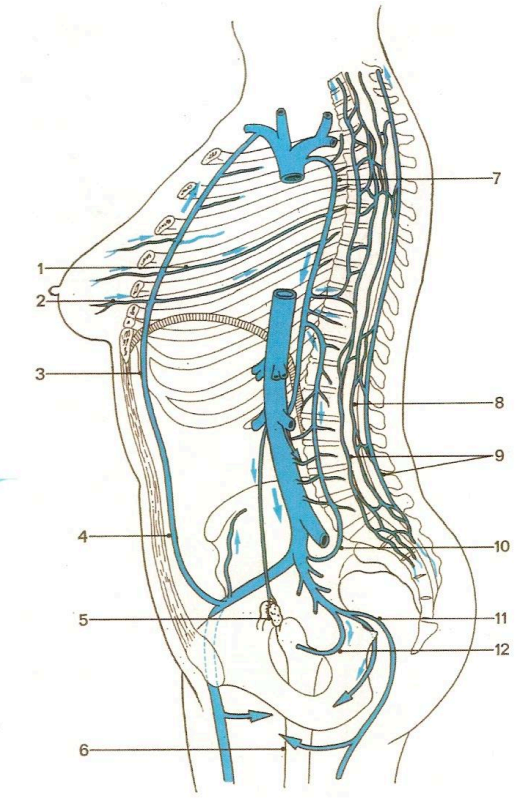


Figure 9. De la veine fémorale à l'abdomen. Référence (36), p 472.

FIG. — Deep venous system

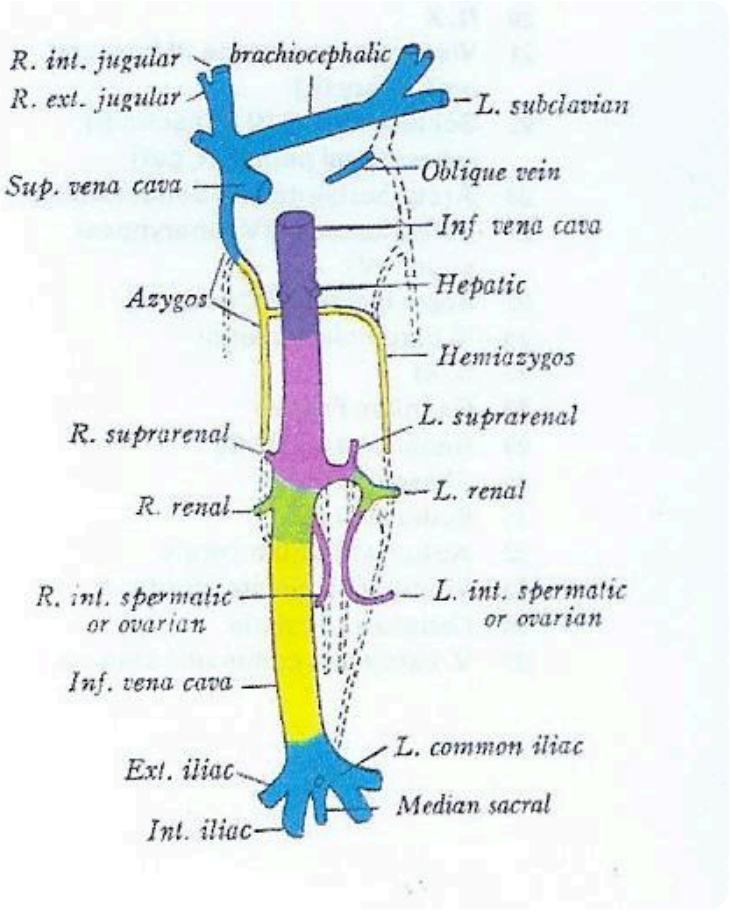


FIG. — Inferior vena cava development

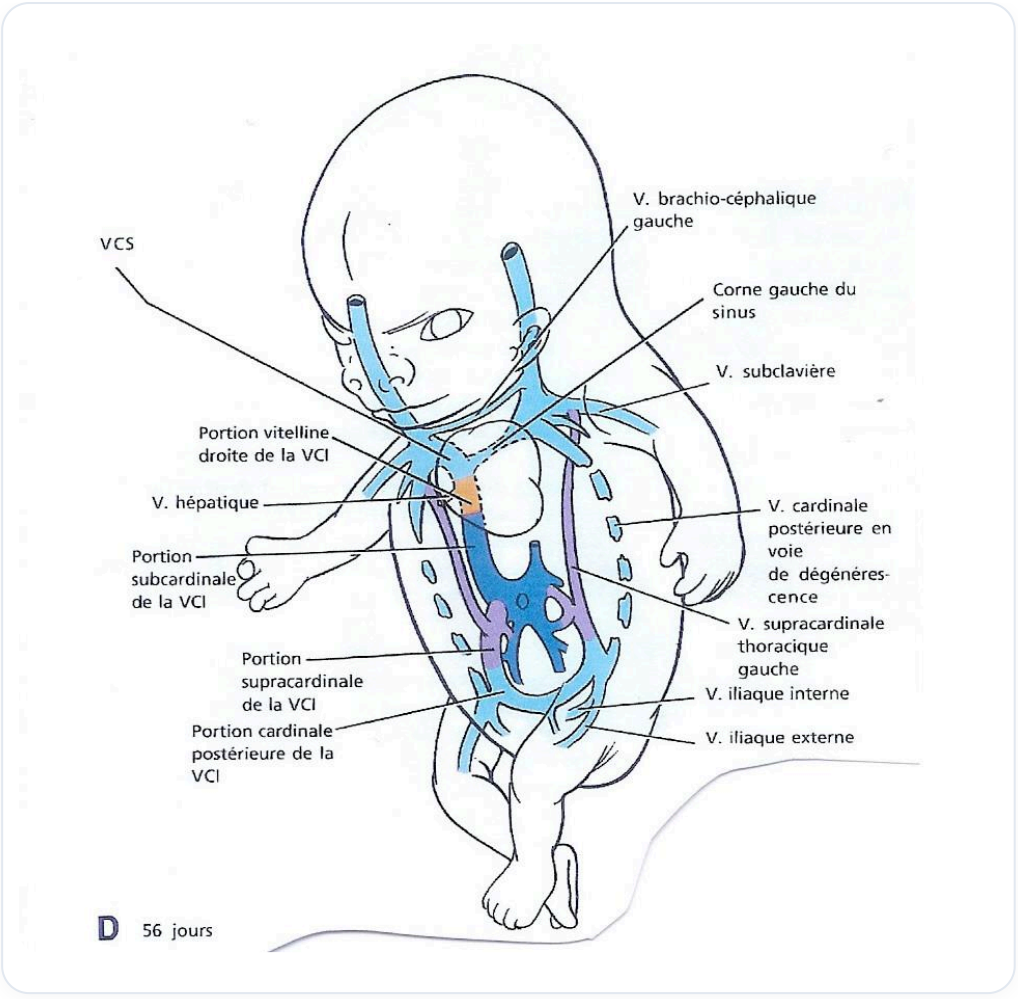


FIG. — Venous System Development